

Software Requirements Specification

For ShopEase — An E-Commerce Website

Prepared as per IEEE 830 SRS Guidelines

Version 1.0

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[Student / Team Name]

[Course / Project Title]

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Table of Contents

- 1. Introduction**
 - 1.1 Purpose
 - 1.2 Document Conventions
 - 1.3 Intended Audience and Reading Suggestions
 - 1.4 Project Scope
 - 1.5 References
- 2. Overall Description**
 - 2.1 Product Perspective
 - 2.2 Product Features
 - 2.3 User Classes and Characteristics
 - 2.4 Operating Environment
 - 2.5 Design and Implementation Constraints
 - 2.6 Assumptions and Dependencies
- 3. System Features (Functional Requirements)**
- 4. External Interface Requirements**
 - 4.1 User Interfaces
 - 4.2 Hardware Interfaces
 - 4.3 Software Interfaces
 - 4.4 Communications Interfaces
- 5. Non-Functional Requirements**
 - 5.1 Performance Requirements
 - 5.2 Security Requirements
 - 5.3 Usability Requirements
 - 5.4 Reliability and Availability
 - 5.5 Scalability and Maintainability
- 6. System Models**
 - 6.1 Use Case Diagram (Description)
 - 6.2 Data Flow Overview
 - 6.3 Entity Relationship Overview
- 7. Other Requirements**
- 8. Appendix A: Glossary**
- 9. Appendix B: Analysis Models**

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for **ShopEase**, a web-based e-commerce platform that allows customers to browse products, place orders, make payments, and track deliveries online, while giving administrators the tools to manage inventory, orders, and customers. This document is intended to serve as the foundation for design, development, and testing of the system, and as an agreement between stakeholders on what the system will and will not do.

1.2 Document Conventions

Requirements in this document are uniquely identified using the prefix **FR** for functional requirements and **NFR** for non-functional requirements, followed by a sequential number (e.g., FR-1, NFR-3). Priority levels are indicated as **High**, **Medium**, or **Low**. The keywords "shall" and "must" indicate mandatory requirements, while "should" indicates a desirable but non-mandatory feature.

1.3 Intended Audience and Reading Suggestions

- **Project Guide / Instructor** — to evaluate scope, completeness, and feasibility.
- **Development Team** — to use as the basis for system design and implementation.
- **Testers / QA** — to derive test cases directly from the stated requirements.
- **Project Team Members** — to maintain a shared understanding of what is being built.

Readers unfamiliar with the project should begin with Section 2 (Overall Description) before proceeding to the detailed requirements in Sections 3–5.

1.4 Project Scope

ShopEase is a responsive e-commerce web application that enables end customers to register, search and browse a catalog of products organized by category, add items to a shopping cart, apply discount coupons, check out securely, and pay online. Registered customers can track order status, view order history, write product reviews, and manage their profile and saved addresses. Administrators can manage the product catalog, inventory levels, categories, orders, customers, and promotional offers through a dedicated admin dashboard. The initial version focuses on a single-vendor store (all products sold by one seller); a multi-vendor marketplace model is out of scope for this release but is listed as a future enhancement.

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications
- Project proposal / synopsis document (internal)
- [Add any textbooks, course material, or online references used]

2. Overall Description

2.1 Product Perspective

ShopEase is a new, self-contained web application. It is not a modification of an existing system. It will interact with an external payment gateway for transaction processing and an email/SMS service for notifications, but otherwise operates as a standalone product with its own database, application server, and web-based front end.

2.2 Product Features

At a high level, the system provides the following major feature groups (each is detailed further in Section 3):

- User registration, login, and profile management
- Product catalog browsing, search, and filtering
- Shopping cart and wishlist management
- Checkout, order placement, and online payment
- Order tracking and order history
- Product ratings and reviews
- Admin dashboard for product, inventory, order, and user management
- Notifications (order confirmation, shipping updates) via email/SMS

2.3 User Classes and Characteristics

User Class	Description
Guest User	Any visitor who can browse the catalog and search products without logging in, but must register/login to purchase.
Registered Customer	A user with an account who can shop, checkout, track orders, write reviews, and manage a personal profile.
Administrator	Manages products, categories, inventory, orders, coupons, and customer accounts through the admin dashboard.
Delivery/Order Manager (optional)	Updates order and shipment status; may be merged with the Administrator role in a smaller deployment.

2.4 Operating Environment

- Client side: any modern web browser (Chrome, Firefox, Edge, Safari) on desktop, tablet, or mobile, supporting a responsive layout.
- Server side: a web/application server (e.g., Node.js/Express, Django, or Spring Boot — to be finalized in the design phase) hosted on a cloud platform or local server.
- Database: a relational database such as MySQL/PostgreSQL (or a NoSQL alternative such as MongoDB, depending on the final technology choice).

2.5 Design and Implementation Constraints

- The system must be developed using technologies approved/available for the project (to be listed once finalized, e.g., HTML/CSS/JavaScript, a chosen backend framework, and a chosen database).
- Online payments must go through a recognized third-party payment gateway; the system will not store raw card details.
- The system must comply with basic data protection practices for storing customer information.
- Development timeline is constrained by the academic project schedule.

2.6 Assumptions and Dependencies

- Users have access to a device with an internet connection and a modern web browser.
- A stable internet connection is available on the server side for payment gateway and notification service calls.
- The third-party payment gateway and email/SMS provider remain available and their APIs stable during development.

3. System Features (Functional Requirements)

This section lists the detailed functional requirements, grouped by feature area. Each requirement is uniquely identified and assigned a priority.

3.1 User Registration and Authentication

Allows visitors to create an account and registered users to securely access the system.

ID	Requirement	Priority
FR-1	The system shall allow a new user to register using name, email address, and password.	High
FR-2	The system shall validate that the email address is unique and properly formatted.	High
FR-3	The system shall allow a registered user to log in using email and password.	High
FR-4	The system shall allow a user to reset a forgotten password via an emailed reset link.	Medium
FR-5	The system shall allow a logged-in user to log out, ending the active session.	High
FR-6	The system should support login via social/OAuth providers (e.g., Google) as an optional convenience.	Low

3.2 Product Catalog, Search, and Browsing

Enables customers to discover products through browsing, category navigation, and search.

ID	Requirement	Priority
FR-7	The system shall display products organized into categories and sub-categories.	High
FR-8	The system shall allow users to search for products by keyword.	High
FR-9	The system shall allow users to filter search/listing results by price range, category, brand, and rating.	Medium
FR-10	The system shall allow users to sort product listings by price, popularity, or newest arrivals.	Medium
FR-11	The system shall display a product detail page with images, description, price, available stock, and customer reviews.	High

3.3 Shopping Cart and Wishlist

Lets customers collect items before purchase or save them for later.

ID	Requirement	Priority
FR-12	The system shall allow a user to add a product (with selected quantity/variant) to the cart.	High
FR-13	The system shall allow a user to update item quantity or remove items from the cart.	High

ID	Requirement	Priority
FR-14	The system shall display the updated cart subtotal whenever the cart contents change.	High
FR-15	The system shall allow a registered user to add products to a wishlist for future purchase.	Medium
FR-16	The system shall persist cart contents for a logged-in user across sessions.	Medium

3.4 Checkout, Orders, and Payment

Covers the process of converting a cart into a confirmed, paid order.

ID	Requirement	Priority
FR-17	The system shall require the user to provide/select a shipping address before checkout.	High
FR-18	The system shall allow the user to apply a valid discount/coupon code at checkout.	Medium
FR-19	The system shall calculate order total including item cost, applicable discounts, taxes, and shipping charges.	High
FR-20	The system shall integrate with a third-party payment gateway to process online payments (cards/UPI/wallets).	High
FR-21	The system shall generate a unique order confirmation upon successful payment and send a confirmation email/notification.	High
FR-22	The system shall reduce product stock quantity automatically once an order is confirmed.	High
FR-23	The system shall allow a user to cancel an order within an allowed time window, subject to order status.	Medium

3.5 Order Tracking and History

Gives customers visibility into past and current orders.

ID	Requirement	Priority
FR-24	The system shall allow a registered user to view a list of all past and current orders.	High
FR-25	The system shall display the current status of an order (e.g., Placed, Shipped, Out for Delivery, Delivered, Cancelled).	High
FR-26	The system shall notify the user by email/SMS when the order status changes.	Medium

3.6 Reviews and Ratings

Allows customers to share product feedback.

ID	Requirement	Priority
FR-27	The system shall allow a customer who has purchased a product to submit a rating (1–5 stars) and a written review.	Medium
FR-28	The system shall display the average rating and all reviews on the product detail page.	Medium

3.7 Administration

Covers back-office capabilities for managing the store.

ID	Requirement	Priority
FR-29	The system shall allow an administrator to add, edit, or remove products and categories.	High
FR-30	The system shall allow an administrator to update stock quantities and view low-stock alerts.	High
FR-31	The system shall allow an administrator to view, update the status of, and manage all customer orders.	High
FR-32	The system shall allow an administrator to view and manage registered customer accounts.	Medium
FR-33	The system shall allow an administrator to create and manage discount coupons and offers.	Medium
FR-34	The system shall provide the administrator with a basic sales/reporting dashboard (e.g., orders per day, revenue, top products).	Low

4. External Interface Requirements

4.1 User Interfaces

- A responsive customer-facing website with: Home page, Category/Listing page, Product Detail page, Cart page, Checkout page, Login/Register pages, User Account/Order History page.
- A separate Admin Dashboard interface for catalog, order, and user management.
- The interface shall follow simple, consistent navigation (header with search and cart icon, footer with policy links) and shall be usable on both desktop and mobile screen sizes.

4.2 Hardware Interfaces

No special hardware is required. The system requires only standard client devices (desktop, laptop, tablet, or smartphone) capable of running a modern web browser, and standard server hardware/cloud infrastructure to host the application and database.

4.3 Software Interfaces

Interface	Purpose
Payment Gateway API (e.g., Razorpay/Stripe/PayPal)	Processing of online payments and refunds.
Email/SMS Service (e.g., SMTP service / SMS API)	Sending order confirmation, status updates, and password-reset messages.
Database System (e.g., MySQL/PostgreSQL/MongoDB)	Persistent storage of users, products, orders, and reviews.

4.4 Communications Interfaces

All client-server communication shall occur over HTTPS. The system shall expose (or consume) RESTful APIs, exchanging data in JSON format, for interactions between the front end, back end, and third-party services.

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-1	Product listing and search pages shall load within 3 seconds under normal load (up to ~100 concurrent users).
NFR-2	The system shall support at least 500 registered users and 10,000 products without significant performance degradation.

5.2 Security Requirements

ID	Requirement
NFR-3	User passwords shall be stored using a strong one-way hashing algorithm; plain-text passwords shall never be stored.
NFR-4	All pages that transmit personal or payment-related data shall use HTTPS/TLS encryption.

ID	Requirement
NFR-5	The system shall not store raw payment card details; card data shall be handled entirely by the PCI-DSS-compliant payment gateway.
NFR-6	The system shall enforce role-based access control so that only administrators can access the admin dashboard.

5.3 Usability Requirements

ID	Requirement
NFR-7	The user interface shall be intuitive enough for a first-time user to complete a purchase without external help.
NFR-8	The website shall be responsive and usable on common screen sizes (mobile, tablet, desktop).

5.4 Reliability and Availability

ID	Requirement
NFR-9	The system shall target at least 99% uptime during business hours (excluding scheduled maintenance).
NFR-10	In case of payment failure, the system shall not deduct stock or mark the order as confirmed, and shall inform the user clearly.

5.5 Scalability and Maintainability

ID	Requirement
NFR-11	The system architecture shall allow new features (e.g., multi-vendor support) to be added with minimal changes to existing modules.
NFR-12	Code shall follow a consistent style/structure and include documentation sufficient for a new team member to understand the codebase.

6. System Models

6.1 Use Case Overview

The primary actors are **Guest**, **Registered Customer**, and **Administrator**. Key use cases include: Register/Login, Browse & Search Products, Manage Cart, Checkout & Pay, Track Order, Write Review (Customer); and Manage Products, Manage Orders, Manage Customers, Manage Coupons (Administrator). A detailed use-case diagram should be inserted here in the final document (e.g., drawn in draw.io/Lucidchart and pasted as an image).

6.2 Data Flow Overview (Level 0)

At the highest level, data flows as follows: the **Customer** interacts with the **Web Application**, which reads/writes product, cart, and order data in the **Database**, and communicates with the external **Payment Gateway** and **Notification Service**. The **Administrator** interacts with the same Web Application through the admin dashboard to update product and order data. A formal DFD diagram should be inserted here.

6.3 Entity Relationship Overview

Core entities are expected to include: **User** (customer/admin), **Product**, **Category**, **Cart** and **CartItem**, **Order** and **OrderItem**, **Payment**, **Review**, and **Coupon**. Key relationships: a User places many Orders; an Order contains many OrderItems, each referencing a Product; a Product belongs to a Category and may have many Reviews. A full ER diagram should be inserted here.

7. Other Requirements

- **Legal:** The system shall include a Privacy Policy and Terms & Conditions page.
- **Internationalization:** Currency and pricing shall initially support a single region/currency (configurable for future expansion).
- **Backup:** The database shall be backed up on a regular schedule to prevent data loss.

8. Appendix A: Glossary

Term	Definition
SRS	Software Requirements Specification
Cart	A temporary collection of products a user intends to purchase
Checkout	The process of confirming an order and completing payment
Coupon	A code that applies a discount to an order
Admin Dashboard	A restricted interface used by administrators to manage the store
Payment Gateway	A third-party service that securely processes online payments

9. Appendix B: Analysis Models

[Insert Use Case Diagram, Data Flow Diagram, and ER Diagram images here in the final submitted version.]